	Republic of the Philippines **Department of Transportation**
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	RE-ENTRY ACTION PLAN (REAP) FORM

INSTRUCTIONS: This form is used in preparing a proposed Re-entry Action Plan in applying to a Scholarship Program. Duly accomplished and signed forms will be submitted to the HRD Division along with the other documentary requirements.

Title of Scholarship Program:	JDS 24th Batch Scholarship Program
Name of the Development Partner	Japan International Cooperation Center (JICE)

I. REAP BACKGROUND AND INFORMATION	
a. REAP Title	**Development of Digital Competency Management System for Philippine Railway Workforce for Sustainable Smart Railway Stations**
b. REAP Objective	To design, develop, and implement a Digital Competency Management System (DCMS) within 12 months at the Philippine Railways Institute (PRI) that will: • Establish a comprehensive digital list of all the skills and competencies needed by railway station personnel to adopt and operate smart stations in the country. • Determine the required skills and competencies for all railway operation personnel working in smart railway station. • Develop a database that monitor all the key skills and competency needed by railway station personnel in smart station.
c. Problem/Opportunity	The Philippine railway sector is entering an important phase of modernization with major projects such as the Metro Manila Subway, North–South Commuter Railway (NSCR), MRT-7 and other railway projects under the Build, Better, More program of the current administration. The upcoming new railway operating lines are being constructed with a design of smart railway station, featuring technologies such as real-time crowd monitoring, automated fare collection (AFC), AI-powered security and surveillance, and integrated passenger information tools. These innovations are designed to make station operations more efficient, reliable, and passenger friendly. Said innovations are opportunity for the Department of Transportation (DOTr) - Philippine Railways Institute (PRI) to

	provide training relevant to the needs of the railway sector and ensuring that railway station personnel possess the required skills and competencies to operate, manage, and maximize the potential of these smart technologies effectively. Currently, PRI's training courses focus primarily on conventional station operations. While these foundational skills remain essential, they are no longer sufficient to meet the demands of smart station environments. Recent initiatives provide clear evidence of this gap. In addition, the PRI Competency Framework Review in 2023 identified emerging skill areas such as digital adaptability, data-informed operations, and system-oriented decision making. This REAP addresses these challenges by developing a Digital Competency Management System (DCMS) tailored for smart railway stations. The DCMS will serve as a centralized digital platform to record, track, and manage the full range of skills and competencies required for modern station operations. It will create a complete digital list of competencies for station personnel, allow PRI specifically the Training Division to monitor personnel progress, and ensure that every member has access to the guidance and resources needed to further develop their capabilities. By linking skills to practical station task, such as handling automated fare collection systems, managing real-time crowd data, and responding to passenger information systems, the DCMS transforms training from a static, manual process into a dynamic, data-informed learning experience. Beyond addressing immediate skill gaps, the system contributes to long-term institutional sustainability. Establishing a centralized, digital framework for smart railway competency development, reduces reliance on external technical assistance, and allows for continuous adaptation of training programs as new technologies and operational challenges emerge. Finally, this initiative is about preparing Philippine railway workforce not only for the stations of today but for smart, efficient, and passenger-focused stations of the future. By implementing the DCMS, the DOTr-PRI can proactively ensure workforce readiness, improve operational efficiency, and support the country's transition to a modern, technology-enabled railway network. This approach strengthens both the people and the institution, ensuring that Philippine railways are prepared for the next generation of smart station operations.
d. Description in relation to the Department i. Organizational Outcome	The primary outcome of this REAP is the development of a Digital Competency Management System (DCMS) that strengthens the skills and competencies of station personnel at the smart railways stations. By providing a centralized, digital platform that records, tracks, and evaluates the competencies required for smart station operations,

	this initiative directly contributes to building workforce competencies. The system ensures that personnel are not only trained in conventional operational tasks but are also equipped with the skills needed to manage modern, technology-enabled stations. Through the DCMS, PRI will be able to map all the competencies required for smart station operations in the Philippines, such as handling automated fare collection systems, monitoring real-time passenger flows, managing AI-powered security tools, and delivering timely passenger information. This allows the organization to identify skill gaps, target training interventions more effectively, and monitor the progress of individual personnel over time. Consequently, staff are better prepared to perform their roles efficiently, safely, and confidently, even as the complexity of station operations increases with the introduction of modern systems. In addition to strengthening individual competencies, this REAP contributes to system development within country's railway sector. The DCMS will serve as an institutional platform that organizes and standardizes skill tracking across all station personnel. This system allows for evidence-based decision-making in workforce planning, training prioritization, and operational readiness assessments. By having a structured and up-to-date digital record of competencies, the railway sector through DOTr-PRI can digitally continuously improve training programs, respond quickly to emerging operational needs, and align human resource capabilities with the evolving demands of smart railway stations. Ultimately, the REAP enhances service delivery indirectly by ensuring that personnel are well-prepared and competent in operating modern stations. With a workforce that possesses the necessary skills for real-time problem solving, passenger service management, and technology driven operations, stations can operate more efficiently, safely, and reliably. The initiative also supports organizational sustainability by establishing a system that can be maintained and updated as technologies evolve, reducing dependence on external technical assistance and fostering long-term institutional resilience. In summary, this REAP addresses building competencies and system development in the host organization. By equipping station personnel with the skills needed for smart station operations and providing a digital platform to manage and track these competencies, the project ensures both workforce readiness and

	organizational capability, directly supporting the delivery of efficient, modern, and passenger-focused railway services.
ii. Target Competency Improvement	The proposed program of study at Yokohama National University, Institute of Urban Innovation will provide and strengthen competencies in digital systems management, smart station operations, and data-driven decision-making. The program offers advanced training in designing and implementing digital frameworks, managing complex urban transport systems, and integrating technology into operational workflows. These competencies are directly relevant to my REAP on developing a Digital Competency Management System (DCMS) for the Philippine railway sector. I believe to improve my ability to analyze and apply international best practices in urban transport and smart mobility to local railway contexts. Learning methodologies for competency assessment, digital tracking, and human-centered system design will enable me to translate theoretical knowledge into practical solutions for workforce training and management. Moreover, exposure to simulation-based and data-informed approaches will help me design training programs that prepare personnel for modern, smart station operations. Overall, this program of study is significant because it equips me with the knowledge and skills necessary to effectively design, develop, and pilot-test the DCMS, ensuring that all railway personnel are prepared for smart railway operations considering the government efforts of digitalization. It will enhance Philippine railway workforce skills and competencies, improve operational readiness, and contribute to the long-term sustainability of modern railway systems in the Philippines.
e. Relation to the Government's Development Plan	The REAP, "**Development of Digital Competency Management System for Philippine Railway Workforce for Sustainable Smart Railway Stations**" directly aligned on the following: • Philippine Development Plan (PDP) 2023–2028, particularly under Chapter 12: Expand and Upgrade Infrastructure development, • Pagtanaw 2050, Digital Transformation and Information and Communications Technology under section 4.11, and • Sustainable Development goals (SDG) under 9.1 Quality, reliable, sustainable, and resilient infrastructure The PDP emphasize the need to modernize the transport sector particularly the railway sector, Pagtanaw 2050 of DOST stating the needs to improve transport infrastructure and public mobility systems that is essential to achieving the Filipino aspirations for a "maginhawang" and "panatag na buhay," also outlined in AmBisyon Natin 2040, and the SDGs, reflecting a country's progress in developing reliable, sustainable, and inclusive infrastructure that

	supports economic productivity and mobility resulted to the growth of passenger volumes across different transport modes, strengthen human resource capabilities, and integrate smart and sustainable technologies to ensure efficient, safe, and inclusive mobility. By developing a DMCS, this project contributes to building a digitally competent and future-ready railway workforce, aligned with the Pagtanaw 2050, SDG, and PDP's. Moreover, this project supports the Build Better More program by ensuring that the upcoming new railway operating lines, like the Metro Manila Subway and the North–South Commuter Railway, are backed by globally competitive railway personnel who have technical and operational skills. By focusing on capacity-building and knowledge transfer, the project helps strengthen the government's broader goal of creating a smart, efficient, and resilient transport system. Ultimately, this initiative contributes to the PDP's vision of a modern, interconnected, and people-centered Philippines where innovation and prepared human capital are key drivers of sustainable national development.
f. Direct Beneficiary	The immediate beneficiaries of this project are the station personnel of the Philippine railway system who will directly interact with and benefit from the Digital Competency Management System (DCMS). This includes train dispatchers, station supervisors, ticketing and customer service staff, and other personnel involved in daily station operations. By providing a clear digital record of the competencies required for smart station operations, the DCMS allows staff to identify skills gaps, track their progress, and access targeted training resources, enabling them to perform their duties more efficiently and confidently. The REAP also indirectly benefits PRI as an institution. By systematizing competency management, PRI can plan training interventions more strategically, improve operational readiness, and reduce reliance on external technical support. This strengthens the organization's capacity to support the smart railway lines, such as the Metro Manila Subway and the North–South Commuter Railway and all existing railway lines that gradually adapting for smart enabled station equipment.

	The target beneficiaries include the station support staff, station personnel across multiple railway lines currently mandated attend DOTr- PRI training courses as per Executive Order 96, series of 2019. Ultimately, the project enhances the competencies of station personnel, improves operational efficiency at stations, and ensures a higher level of service for passengers, contributing to a safer, smarter, and more reliable railway system in the Philippines.
g. Gender Equality, Disability, and Social Inclusion (GEDSI)	The implementation of this project will contribute to creating responsive and inclusive railway stations, ensuring that women and other marginalized groups such as Indigenous Peoples (IP), Persons with Disabilities (PWD), and members of the LGBTQ+ community will benefit from improved safety, accessibility, and service quality. By developing DCMS, station personnel will receive targeted training that emphasizes inclusive practices, crowd management, and customer service tailored to the needs of diverse passengers. Specific strategies include are the training of staff to assist passengers with mobility challenges, ensuring proper use of ramps, elevators, and accessible ticketing systems, and providing clear, real-time passenger information to help all commuters navigate stations safely and efficiently. Women and LGBTQ+ passengers will benefit from heightened awareness and responsiveness to safety concerns, including proper handling of harassment reports and maintaining secure waiting areas. Indigenous and other marginalized communities will benefit from personnel trained in culturally sensitive communication, signage in multiple languages where possible, and equitable service delivery. Through these initiatives, the REAP not only builds technical competencies but also instills inclusive, human-centered practices in station operations. As a result, marginalized groups can travel more easily, safely, and confidently, reinforcing the role of Philippine railway stations as safe, accessible, and responsive spaces for all passengers.
h. Output	**Expected Outputs of the REAP:** • **Digital Competency Management System (DCMS):** A centralized platform to record, track, and manage the skills and competencies of station personnel for smart railway operations. • **Digital Competency Framework:** A comprehensive list of technical, operational, and customer service skills required for effective smart station management. • **Training Curricula:** Targeted training content aligned with identified skill gaps, focusing on practical application for day-to-day smart station operations. • **Reports and Analytics for future Policy Research to develop Human Resource Competency:** Tools to monitor personnel progress, assess training effectiveness, and support evidence-based workforce planning for future railway policy research.

	The DCMS enables the railway sector to systematically monitor and enhance workforce capabilities, ensuring staff are ready to manage smart stations efficiently. The competency framework and training modules provide structured guidance for skill development, reducing operational errors and improving passenger service. Also the reports and analytics allow for data-driven decision-making in training and human resource management and knowledge transfer ensures sustainability by building institutional capacity to maintain and continuously update competencies. finally, these outputs strengthen personnel competencies, enhance operational readiness, and develop organizational systems, enabling station personnel to efficiently manage smart railway stations and support the long-term sustainability of the Philippine railway network.
II. REAP TIMELINE AND IMPLEMENTATION	
a. Measurability	The success of this REAP will be measured by how effectively the Digital Competency Management System (DCMS) helps station personnel develop the skills they need to operate modern, smart stations. We will track progress through both numbers and real-life impact, how many personnel have their competencies fully recorded in the system, how many complete the targeted training, and how their skills improve through pre- and post-training assessments and practical activity. Success will also be reflected in how confidently staff handle smart station operations, respond to passenger needs, and apply new knowledge in day-to-day situations. Regular feedback from personnel and supervisors will provide insight into what is working well and what needs adjustment. **Milestones** • **Months 1–2:** Finalize the digital competency framework, identifying all skills needed for smart station operations. • **Months 3–5:** Develop the DCMS platform and pilot it with a small group of station personnel. • **Months 6–8:** Roll out targeted training sessions and integrate them into the DCMS. • **Months 9–10:** Expand DCMS use to all station personnel and

	monitor how well it is being adopted. • **Months 11–12:** Conduct post-training assessments, gather feedback, and review operational improvements to ensure the system is fully effective and sustainable. By combining practical skills assessments, system usage data, and staff feedback, this REAP will ensure that personnel are better prepared, more confident, and ready to operate smart, responsive stations that serve passengers efficiently and safely.
b. Sustainability	To ensure sustainability, the Digital Competency Management System (DCMS) will be designed as a permanent, integrated tool within the Philippine Railways Institute (PRI). This includes building internal capacity so that staff can continuously update, monitor, and manage competencies without relying on external support. This includes but not limited to: • Standard Operating Procedures (SOPs): Establishing clear guidelines for updating competencies, tracking training progress, and integrating the system into daily operations. • Periodic Reviews and Updates: Scheduling regular evaluations of the DCMS to ensure it stays aligned with evolving smart station technologies and operational needs. • Knowledge Sharing and Documentation: Capturing lessons learned, best practices, and process manuals to maintain institutional memory and facilitate onboarding of future personnel. • Integration into Organizational Strategy: Embedding competency management into PRI's long-term workforce development and smart station readiness plans. • Training of Trainers (ToT): Equipping selected PRI personnel with the knowledge and skills to manage the DCMS, conduct training, and mentor new staff. By building internal ownership, formalizing processes, and linking the DCMS to the organization's broader goals, the REAP ensures that improvements in staff skills and operational readiness become a sustainable part of DOTr-PRI's operations, supporting the long-term development of smart, responsive, and efficient railway stations.

c. Predictive Action Steps with Identified Success Indicators

i. **Phase/Stage**	ii. **Expected Output**	iii. **Timeline**
Phase 1: Competency Framework Development	Completed digital competency framework identifying all skills required for smart station operations. Success indicator: Framework reviewed and approved by PRI division chiefs.	Months 1–2
Phase 2: DCMS Design and Pilot Testing	Functional DCMS platform tested with a small group of station personnel. Success indicator: Positive feedback from pilot users; at least 80% system functionality verified.	Months 3–5

Phase 3: Training Module Integration	Training content and guidelines integrated into the DCMS. Success indicator: Personnel complete at least 70% of assigned modules during pilot training sessions.	Months 6–8
Phase 4: Full Rollout and Adoption	DCMS fully implemented across all relevant station personnel. Success indicator: 90% of target staff actively using the system and tracking competency progress.	Months 9–10
Phase 5: Post-Implementation Assessment and Review	Final evaluation of competency improvement, system utilization, and operational readiness. Success indicator: Demonstrate improvement in staff skills, reduced operational errors, and increased confidence in managing smart station systems.	Months 11–12
d. Target Date to Start	After JDS Program (Estimate November 2028)	
e. Target Date to End	November 2029	

III. RESOURCES

1. Software Development & DCMS Platform
Development, customization, and testing of the Digital Competency Management System.
Cost Php 120,000
    - Funding Source: The project will utilize internal budget support from PRI's Research and Development and Policy Formulation budget.

2. Training Modules & Materials
Creation of digital and printed training materials, guides, and handbooks for station personnel.
Cost Php 50,000
    - Funding Source: The project will utilize internal budget support from PRI's Research and Development and Policy Formulation budget.

3. Workshops / Knowledge Transfer Sessions
Conducting training sessions, including venue, logistics, and facilitator fees for PRI staff.
Cost Php 40,000
    - Funding Source: The project will utilize internal budget support from PRI's Research and Development and Policy Formulation budget.

4. Assessment Tools & Analytics
Pre- and post-training assessments, data collection, and monitoring tools for evaluating staff competencies.

Cost Php 25,000 • Funding Source: The project will utilize internal budget support from PRI's Research and Development and Policy Formulation budget. 5. Contingency Fund Miscellaneous expenses such as minor software updates, printing, and unforeseen operational costs. Cost Php 15,000 • Funding Source: The project will utilize internal budget support from PRI's Research and Development and Policy Formulation budget.	
IV. COMMENT(S) AND RECOMMENDATION(S)	
a. Immediate Superior (Mentor)	
b. Head of the Office/Division/Service	
c. Human Resource Development Division	

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